

## ROYAL ISLAND, ON, Canada

WMO: 718590

Lat: 49.467N

Lon: 94.761W

Elev: 1079

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.6      | -21.1 | -32.0                         | 0.9 | -25.3 | -26.9 | 1.3 | -20.8 | 20.9                  | 11.6 | 18.5 | 8.1  | 5.0                   | 270  | 0.541 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.3                   | 83.8            | 70.2 | 81.1 | 68.3 | 78.7 | 67.1 | 73.1                | 80.1 | 71.1 | 77.9 | 69.2 | 75.9 | 5.5                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.8                            | 118.2 | 77.3 | 68.8 | 110.1 | 75.3 | 66.9 | 102.8 | 73.2 | 37.6          | 80.2 | 35.7 | 77.8 | 34.1 | 75.9 | 82.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 20.5          | 17.8 | 15.5 | -32.9                      | 89.0 | 5.0                | 3.1 | -36.5                                              | 91.2 | -39.4      | 93.0 | -42.2      | 94.7 | -45.8      | 96.9 |    |    |
| (5)               |      |      | -32.7                      | 76.4 | 5.2                | 2.9 | -36.4                                              | 78.4 | -39.4      | 80.1 | -42.3      | 81.7 | -46.1      | 83.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 39.6        | 4.2   | 8.0   | 23.4  | 38.8  | 52.9  | 63.7  | 69.7  | 68.1  | 59.3  | 44.6  | 28.4  | 11.4  |  |  |
|                                                                        | DBStd       | 25.12       | 14.57 | 13.72 | 13.33 | 10.03 | 8.35  | 5.90  | 4.97  | 5.33  | 7.41  | 8.35  | 10.82 | 13.69 |  |  |
|                                                                        | HDD50       | 5907        | 1419  | 1177  | 824   | 353   | 66    | 1     | 0     | 0     | 12    | 211   | 648   | 1196  |  |  |
|                                                                        | HDD65       | 9663        | 1884  | 1597  | 1288  | 787   | 382   | 92    | 15    | 30    | 198   | 632   | 1097  | 1661  |  |  |
|                                                                        | CDD50       | 2095        | 0     | 0     | 1     | 16    | 156   | 411   | 611   | 562   | 292   | 45    | 1     | 0     |  |  |
|                                                                        | CDD65       | 376         | 0     | 0     | 0     | 0     | 7     | 52    | 161   | 127   | 28    | 1     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1988        | 0     | 0     | 0     | 0     | 53    | 265   | 888   | 657   | 124   | 1     | 0     | 0     |  |  |
|                                                                        | CDH80       | 354         | 0     | 0     | 0     | 0     | 4     | 26    | 175   | 131   | 18    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.8         | 6.7   | 6.5   | 6.8   | 7.4   | 7.5   | 6.0   | 5.4   | 5.5   | 6.4   | 8.1   | 8.6   | 6.9   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 27.4        | 1.2   | 0.9   | 1.3   | 1.5   | 3.2   | 3.9   | 4.1   | 3.0   | 3.2   | 2.3   | 1.6   | 1.2   |  |  |
|                                                                        | PrecMax     | 34.1        | 2.2   | 1.9   | 2.3   | 2.8   | 6.2   | 8.1   | 7.9   | 6.3   | 6.0   | 5.0   | 4.0   | 2.2   |  |  |
|                                                                        | PrecMin     | 18.1        | 0.4   | 0.2   | 0.4   | 0.6   | 1.2   | 1.9   | 1.5   | 0.2   | 1.0   | 0.4   | 0.5   | 0.5   |  |  |
|                                                                        | PrecStd     | 4.5         | 0.4   | 0.4   | 0.6   | 0.6   | 1.5   | 1.7   | 1.6   | 1.5   | 1.5   | 1.1   | 0.9   | 0.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 37.3  | 41.8  | 56.7  | 69.6  | 79.3  | 83.2  | 88.2  | 88.0  | 82.5  | 70.9  | 55.5  | 39.2  |  |  |
|                                                                        |             | MCWB        | 34.5  | 37.1  | 47.9  | 52.1  | 60.6  | 68.0  | 74.1  | 72.9  | 69.0  | 58.8  | 48.0  | 35.4  |  |  |
|                                                                        | 2%          | DB          | 32.8  | 37.2  | 48.5  | 63.3  | 74.7  | 79.8  | 84.2  | 83.7  | 77.4  | 65.4  | 49.1  | 34.6  |  |  |
|                                                                        |             | MCWB        | 31.5  | 34.4  | 40.7  | 49.3  | 59.5  | 66.2  | 71.4  | 69.6  | 66.3  | 56.3  | 43.9  | 32.5  |  |  |
|                                                                        | 5%          | DB          | 29.2  | 33.3  | 44.0  | 58.0  | 70.6  | 77.2  | 81.6  | 80.4  | 73.7  | 60.7  | 45.2  | 32.4  |  |  |
|                                                                        |             | MCWB        | 28.0  | 31.6  | 38.6  | 46.2  | 57.3  | 65.2  | 69.6  | 67.8  | 63.6  | 53.5  | 41.4  | 31.0  |  |  |
|                                                                        | 10%         | DB          | 25.1  | 29.4  | 40.1  | 53.6  | 66.3  | 74.1  | 79.0  | 77.7  | 70.2  | 56.2  | 42.1  | 29.8  |  |  |
|                                                                        |             | MCWB        | 24.1  | 27.4  | 35.6  | 43.8  | 55.6  | 63.8  | 68.2  | 66.5  | 62.2  | 50.6  | 39.2  | 28.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 34.9  | 38.4  | 48.5  | 54.0  | 67.1  | 73.0  | 77.0  | 75.7  | 71.7  | 61.3  | 49.2  | 35.9  |  |  |
|                                                                        |             | MCDB        | 37.1  | 40.9  | 55.5  | 67.1  | 73.7  | 79.0  | 84.1  | 83.5  | 79.5  | 65.9  | 53.9  | 38.6  |  |  |
|                                                                        | 2%          | WB          | 31.6  | 34.5  | 42.5  | 50.4  | 62.8  | 69.5  | 73.9  | 72.4  | 68.3  | 57.4  | 44.9  | 33.0  |  |  |
|                                                                        |             | MCDB        | 32.7  | 36.6  | 47.2  | 59.6  | 69.4  | 76.1  | 80.7  | 80.0  | 74.0  | 64.6  | 48.3  | 34.3  |  |  |
|                                                                        | 5%          | WB          | 28.1  | 31.8  | 38.9  | 47.7  | 59.4  | 67.1  | 71.8  | 70.1  | 65.6  | 54.4  | 41.9  | 30.9  |  |  |
|                                                                        |             | MCDB        | 29.3  | 33.4  | 43.6  | 56.4  | 66.1  | 73.8  | 78.4  | 77.7  | 71.6  | 59.7  | 44.8  | 32.2  |  |  |
|                                                                        | 10%         | WB          | 24.1  | 27.4  | 36.0  | 44.7  | 57.0  | 65.2  | 70.1  | 68.2  | 63.4  | 50.9  | 39.1  | 28.5  |  |  |
|                                                                        |             | MCDB        | 25.0  | 29.1  | 39.4  | 53.0  | 64.3  | 71.5  | 76.5  | 74.7  | 68.8  | 55.3  | 42.0  | 29.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 17.8  | 21.3  | 20.7  | 18.7  | 16.7  | 15.3  | 15.3  | 15.3  | 14.0  | 11.2  | 10.0  | 14.1  |  |  |
|                                                                        |             | MCDBR       | 16.6  | 18.0  | 20.2  | 24.1  | 22.3  | 19.3  | 18.1  | 18.9  | 18.5  | 18.0  | 13.3  | 13.7  |  |  |
|                                                                        | 5% WB       | MCWBR       | 15.2  | 14.6  | 13.9  | 13.2  | 10.3  | 9.4   | 8.4   | 8.3   | 9.1   | 10.2  | 8.7   | 11.8  |  |  |
|                                                                        |             | MCWBR       | 16.3  | 15.5  | 18.8  | 22.1  | 17.8  | 16.8  | 16.4  | 16.7  | 16.9  | 16.6  | 12.2  | 13.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.247 | 0.259 | 0.293 | 0.332 | 0.349 | 0.372 | 0.378 | 0.369 | 0.346 | 0.303 | 0.271 | 0.246 |  |  |
|                                                                        |             | taud        | 2.326 | 2.306 | 2.377 | 2.434 | 2.455 | 2.411 | 2.420 | 2.465 | 2.507 | 2.590 | 2.485 | 2.349 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 261   | 285   | 293   | 291   | 289   | 281   | 278   | 276   | 272   | 269   | 252   | 246   |  |  |
|                                                                        |             | Edh at Noon | 21    | 28    | 31    | 33    | 34    | 36    | 35    | 32    | 28    | 21    | 18    | 18    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 413   | 782   | 1162  | 1508  | 1706  | 1783  | 1889  | 1567  | 1153  | 667   | 359   | 288   |  |  |
|                                                                        | RadStd      | RadStd      | 43    | 61    | 113   | 148   | 121   | 145   | 103   | 119   | 99    | 95    | 46    | 32    |  |  |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   |       |       |
| (47) Regional (22 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   |       |       |

Nomenclature: See separate page